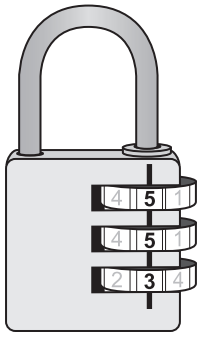
16 There are three dials on a combination lock.
Each dial can be set to one of the numbers 1, 2, 3, 4, 5
The three digit number 553 is one way the dials can be set, as shown in the diagram.



(a) Work out the number of different three digit numbers that can be set for the combination lock.

(a) Five choices for each of three dials:

5 × 5 × 5 = 125 combinations

1 mark
Multiply five choices three times.

1 mark
125 combinations.

125 **(2)**

(b) How many of the possible three digit numbers have three different digits?

(b) No repeats: 5 then 4 then 3 choices.

5 × 4 × 3 = 60 combinations

1 mark
Different digits reduce the choices.

1 mark
60 combinations.

60

(2)

(Total for Question 16 is 4 marks)

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